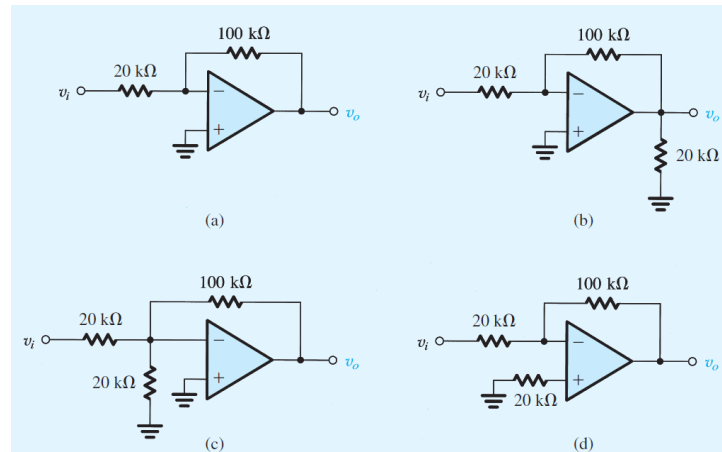
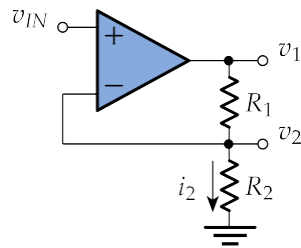


Homework 5 Topic: Op-amps

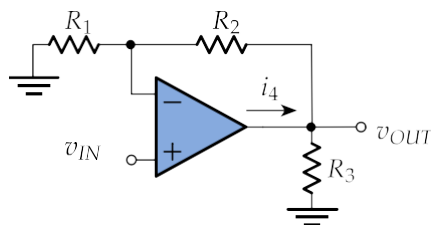
- Assuming ideal op amps, find the voltage gain v_o/v_i and input resistance R_{in} of each of the circuits below:



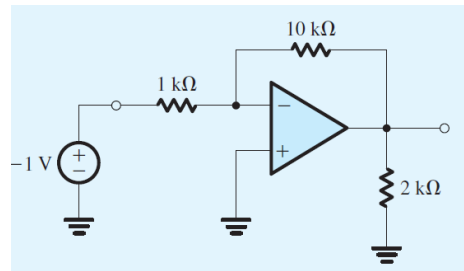
- Find v_1 , v_2 , and i_2 in terms of v_{IN} (and the resistors).



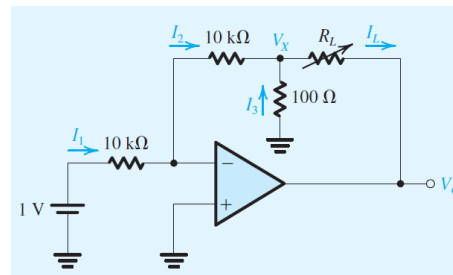
- Find v_{OUT} and i_4 in terms of v_{IN} (and the resistors).



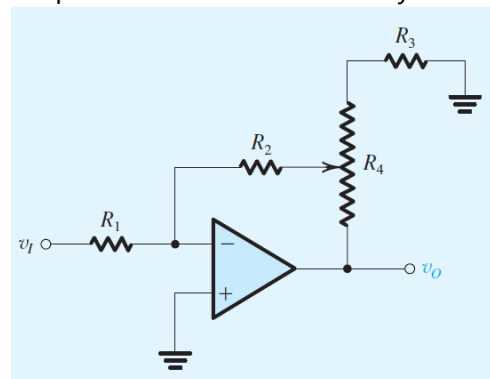
- For the circuit below, assuming an ideal op amp, find the currents through all branches and the voltages at all nodes. Since the current supplied by the op amp is greater than the current drawn from the input signal source, where does the additional current come from?



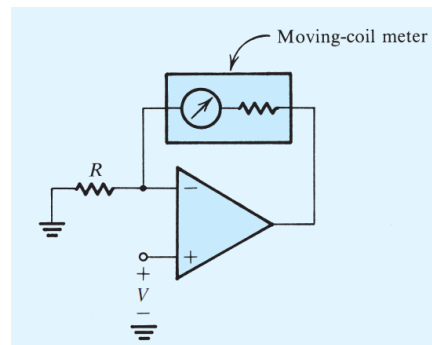
- 5.
6. The circuit in below utilizes an ideal op amp.
- Find I_1 , I_2 , I_3 , I_L , and V_x .
 - If V_O is not to be lower than -13 V, find the maximum
 - allowed value for R_L .
 - If R_L is varied in the range $100\ \Omega$ to $1\text{ k}\Omega$, what is the corresponding change in I_L and in V_O ?



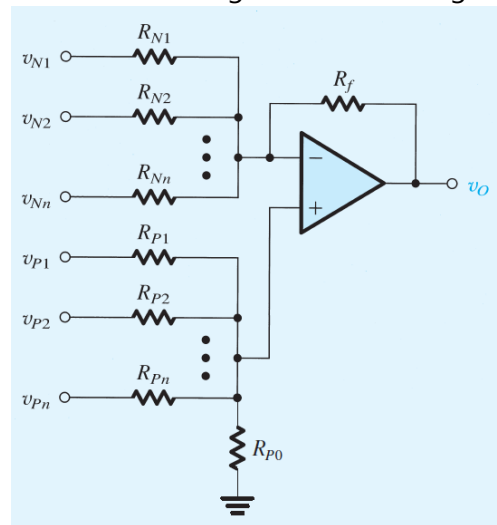
7. Design the circuit shown in Fig. P2.35 to have an input resistance of $100\text{ k}\Omega$ and a gain that can be varied from -1 V/V to -100 V/V using the $100\text{-k}\Omega$ potentiometer R_4 . What voltage gain results when the potentiometer is set exactly at its middle value?



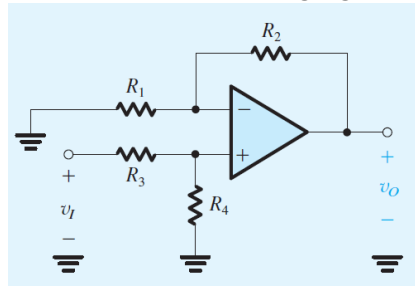
8. Figure below shows a circuit for an analog voltmeter of very high input resistance that uses an inexpensive moving-coil meter. The voltmeter measures the voltage V applied between the op amp's positive-input terminal and ground. Assuming that the moving coil produces full-scale deflection when the current passing through it is $100\ \mu\text{A}$, find the value of R such that a full-scale reading is obtained when V is $+10$ V. Does the meter resistance shown affect the voltmeter calibration?



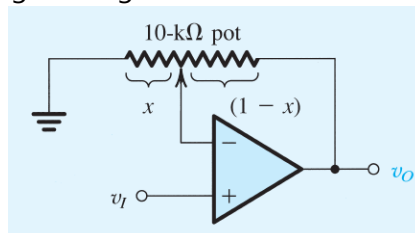
9. Design a circuit, using one ideal op amp, whose output is $v_O = v_{I1} + 2v_{I2} - 9v_{I3} + 4v_{I4}$. (Hint: Use a structure similar to that shown in general form in figure below)



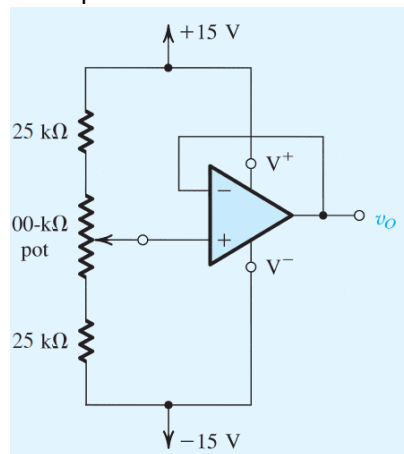
10. Derive an expression for the voltage gain, v_O/v_I , of the circuit below



11. The circuit below utilizes a 10-k Ω potentiometer to realize an adjustable-gain amplifier. Derive an expression for the gain as a function of the potentiometer setting x . Assume the op amp to be ideal. What is the range of gains obtained? Show how to add a fixed resistor so that the gain range can be 1 to 11 V/V. What should the resistor value be?

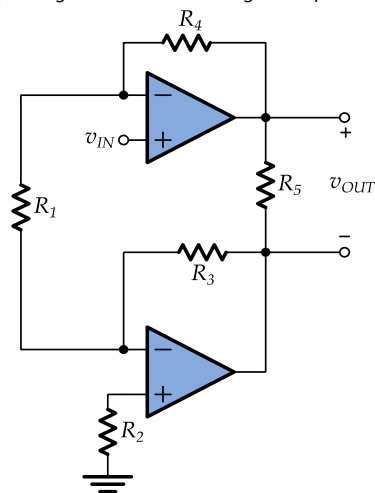


12. Figure P2.59 shows a circuit that provides an output voltage v_O whose value can be varied by turning the wiper of the 100-k Ω potentiometer. Find the range over which v_O can be varied. If the potentiometer is a "20-turn" device, find the change in v_O corresponding to each turn of the pot.

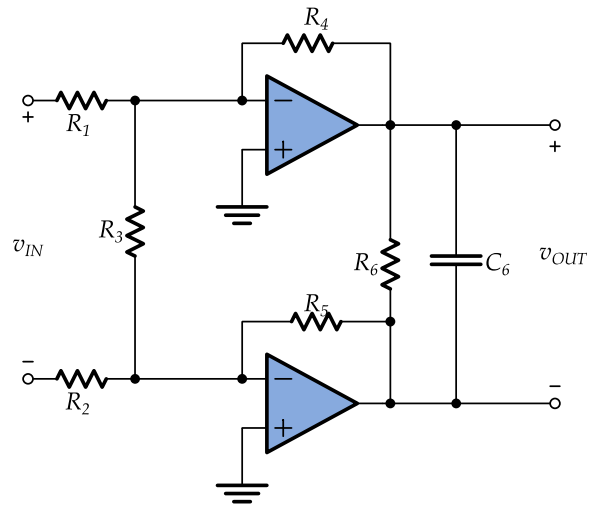


13. Find the gain for the following circuits, assuming ideal op-amps.

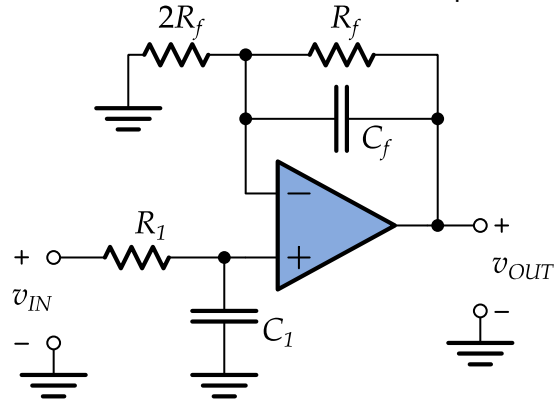
- a. For this circuit, $R_1 = R_2 = R_5 = 2 \text{ k}\Omega$ and $R_3 = R_4 = 1 \text{ k}\Omega$.



- b. For this circuit, $R_1 = 1 \text{ k}\Omega$, $R_2 = 5 \text{ k}\Omega$, $R_3 = 2 \text{ k}\Omega$, $R_4 = 3 \text{ k}\Omega$ and $R_5 = 15 \text{ k}\Omega$, $R_6 = 6 \text{ k}\Omega$, and $C_6 = 1 \mu\text{F}$.



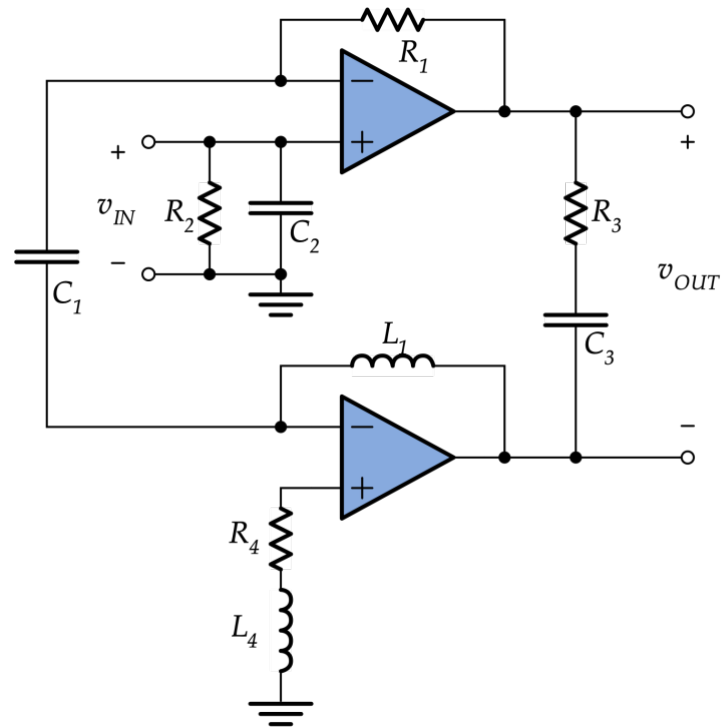
14. Find the transfer function $H(j\omega) = V_{out}/V_{in}$ for the op amp circuit below, assuming an ideal op amp. Sketch the Bode magnitude and phase plots using straight-line approximations. You will have to come up with values for the resistors and capacitors.



15. Assuming ideal op-amps,

- Find the transfer function, $H(\omega) = V_{out}/V_{in}$
- Find the number of poles and zeros for this transfer function
- For extra credit, find the values of the poles and zeros for this transfer function

Use $R_1=1\text{ k}\Omega$, $R_2=2.2\text{ k}\Omega$, $R_3=3.3\text{ k}\Omega$, $R_4=4.7\text{ k}\Omega$, $C_1=2.2\text{ }\mu\text{F}$, $C_2=0.1\text{ mF}$, $C_3=0.01\text{ mF}$, $L_1=3.3\text{ mH}$, and $L_4=39\text{ }\mu\text{H}$.



16. Design a circuit that meets the following specifications using only parts from the list below (you do not have to use all of the parts).

- Gain is $20 \text{ dB} \pm 6 \text{ dB}$ for $\omega \geq 100 \text{ rad/sec}$
- Gain is $0 \text{ dB} \pm 6 \text{ dB}$ for $\omega < 10 \text{ rad/sec}$
- $R_{in} > 150 \text{ k}\Omega$

Available parts:

- (10) resistors of any value $< 10 \text{ k}\Omega$; • (5) ideal op-amps; and
- any capacitors.

17. Assuming ideal op-amps,

- Find the voltage gain, v_{OUT}/v_S
- Find the input resistance seen by the source

Use $R_5=6.9 \text{ k}\Omega$, $R_1=3.3 \text{ k}\Omega$, $R_2=4.7 \text{ k}\Omega$, $R_3=2.2 \text{ k}\Omega$, $R_4=75 \text{ k}\Omega$, $R_5=19 \text{ k}\Omega$, $R_6=27 \text{ k}\Omega$, $R_7=86 \text{ k}\Omega$, $R_8=10 \text{ k}\Omega$, and $R_L=5.6 \text{ k}\Omega$.

